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Cardiac cycle timing and contractility following acute sport-related concussion

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ABSTRACT

Cardiac sequelae following sport-related concussion are not well understood. This study describes changes in the cardiac cycle timing intervals and contractility parameters during the acute phase of concussion. Twelve athletes (21 ± 2 years, height = 182 ± 9 cm, mass = 86 ± 15 kg, BMI = 26 ± 3 kg/m^2) were assessed within 5 days of sustaining a diagnosed concussion against their own pre-season baseline. A non-invasive cardiac sensor (LLA Recordis™) was used to record the cardiac cycle parameters of the heart for 1 minute during supine rest. Cardiac cycle timing intervals (Isovolumic relaxation and contraction time, Mitral valve open to E wave, Rapid ejection period, Atrial systole to mitral valve closure, Systole, and Diastole) and contractile forces (Twist force and Atrial systole: AS) were compared. Systolic time significantly decreased during acute concussion ($p = 0.034$). Magnitude of AS significantly increased during acute concussion ($p = 0.013$). These results imply that concussion can result in altered systolic function.

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Introduction

Competitive play during contact sports can result in mild traumatic brain injury (mTBI) or sport-related concussion (SRC), with American football, men's wrestling, and men's and women's ice hockey having the high prevalence of injury in the NCAA (Zuckerman et al., 2015). Both acute and chronic concussions can alter autonomic cardiac function (Krzyzaniak & Fatehi Hassanabad, 2021), and research has shown the direct influence of SRC on heart rate variability (HRV; S. A. Bishop et al., 2018; Charron et al., 2021) and baroreceptor sensitivity (BRS; La Fountaine et al., 2019). What is not known is how SRC can impact the mechanics of the heart, specifically, the cardiac cycle timing and contractile parameters. Considering that baseline assessments for the majority of athletes in competitive sport do not typically include collection of cardiac cycle timing intervals, such information would be an asset in understanding the mechanisms of cardiac dysfunction in SRC and its sequelae (Krzyzaniak & Fatehi Hassanabad, 2021).

Cardiac cycle timing parameters can provide vital information about overall heart health and function. These parameters are represented by the Wiggers Diagram and comprise periods of systole and diastole (Wiggers, 1915). With echocardiographic data,

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ejection time can be calculated (systolic time), as well as the isovolumic contraction (IVCT) and relaxation (IVRT) timing intervals (Biering-Sorensen et al., 2016). Despite the great potential to understand cardiac pathology by measuring the timing intervals, there is still much unknown pertaining to what the normal values are expected to be in SRC.

Previous research using echocardiography has attempted to differentiate normal and pathological cardiac time intervals in the general population and found that these values differed depending on age and sex (Biering-Sorensen et al., 2016). Furthermore, these parameters can be used to calculate cardiac performance indices, including systolic (IVCT/systolic time; SPI), diastolic (IVRT/systolic time; DPI), and the heart or myocardial [(IVRT+IVCT)/systolic time; HPI)] (Tei, 1995) performance indices. Major adverse cardiac events such as hospital admission due to ischaemic heart disease or heart failure can also alter these parameters in comparison to those with no cardiac history (Biering-Sorensen et al., 2015). Thus, while these cardiac parameters are well understood in a healthy population and those with a history of cardiac complications, a better understanding of alterations in a concussed state can help shed light on the mechanism of cardiac impairments in SRC, and potentially help for recovery from injury. Use of echocardiograms can be demanding on resources and time. Therefore, there is a requirement for a more time-efficient assessment of cardiac function in acute concussion, and better understanding of cardiac cycle timing intervals for information pertaining to the implications of cardiac autonomic dysfunction in SRC.

The purpose of this novel study was to characterize changes in cardiac function in participants who present with SRC. Based on our previous research (Neary et al., 2020), we hypothesized that concussion would alter cardiac function and mechanics negatively as a whole, including impairments in timing intervals such as systole.

Methods

Participants

This study was approved by the University of Regina Ethics Research Board (REB#55R1213). Participants signed an informed consent form prior to beginning the study. One-hundred and fifty-two males and 93 females were recruited for baseline assessment during their pre-season. The male athletes were competing in football, swimming, hockey, or basketball. The female athletes were competing in soccer, basketball, hockey, volleyball, or swimming. Of these participants, 12 (21 ± 2 years of age, height = 182 ± 9 cm, mass = 86 ± 15 kg, BMI = 26 ± 3 kg/m^2) were followed up due to sustaining SRC. Participants refrained from alcohol intake (24 hours), caffeine intake (6 hours) and exercise (6 hours; Neary et al., 2019). All participants were university, varsity-level athletes. The Sport Concussion Assessment Tool (SCAT) symptom scale was completed prior to beginning the cardiac assessment (Echemendia et al., 2017). Participants completed a baseline assessment and repeated the protocol if they suffered from a concussion, as diagnosed by the team physician following initial assessment completed by a certified athletic therapist.

Data Collection

A non-invasive cardiac sensor (LLA Recordis™; LLA Technologies, Langley, BC, Canada) was used to record the cardiac cycle timing and force production parameters of the heart (Singh et al., 2021). The sensor was snapped into a single adhesive ECG gel-electrode and placed 1 cm above the xiphoid process on the sternum of the chest over the skin. Participants were asked to rest in a supine position, with a pillow under the head for comfort, for ~1 minute prior to turning the cardiac sensor on. The cardiac data were collected for 1 minute. This method was repeated during every follow-up data collection period post-concussion. Data collection was completed at the same time of the day as the baseline assessment during the follow-up testing to avoid circadian changes.

Data analysis

The analysis of the cardiac signal collected at the sternum has been reported elsewhere (Singh et al., 2021). Briefly, an in-house, independent (proprietary) algorithm (LLA Technologies Inc.) assessed the extracting cardiac features, and all noise signals were removed from the signal. Specifically, data were collected at 500 Hz and a first-order Butterworth bandpass filter with a low cut-off frequency of 1 Hz and a high cut-off frequency of 30 Hz filter was applied to smooth the signal and adjust for the baseline wandering. Fiducial points of the cardiac cycle included the following: mitral valve closure (MVC), aortic valve opening (AVO), aortic twist (ATT), aortic systole, REP, aortic valve closure (AVC), ventricular untwisting, and mitral valve opening (MVO). These fiducial points translated to temporal features, being diastole (MVC – MVO timing), systole (AVO – AVC timing), IVCT (MVC – AVO), IVRT (AVC – MVO), atrial systole to mitral valve closure (AS to MVC), mitral valve open to E wave (MVO to E), and end of rapid ejection (REP). Heart rate (AVOn+1 – AVOn) was calculated in beats per minute. Twist force (TF), or the magnitude of the cardiac contraction acceleration (milligravity, mG) at the sternum, was used to assess contractility, and AS was used to assess atrial contractility (Neary et al., 2011).

Statistical analyses

A within-subject statistical model was used so that the concussed participants were compared to their own baseline data only. Considering the small sample size, a Wilcoxon signed-rank test was used to compare changes from baseline to with a follow-up period of within 5 days of injury. Cardiac cycle data are presented as median [interquartile range (IQR)].

Results

Participants were followed up at 3 ± 1 day's post-injury. Fifty per cent of the participants compete in male American football, and the remaining 50% compete in female soccer, male ice hockey, male basketball, or female ice hockey. No participants lost consciousness upon sustaining the injury. During the acute period follow-up, the mean sum of the SCAT score was 12 ± 10.

Acute concussion results for the cardiac cycle parameters are presented in Table 1. Twelve participants were included in the group analyses. A significant decrease in systolic time ($Z = -2.12$, $p = 0.034$) from 298 ms (IQR = 42 ms) at baseline to 276 ms (IQR = 32 ms) during acute concussion and a significant increase ($Z = -2.47$, $p = 0.013$) in AS from 10 mG (IQR = 6 mG) at baseline to 13 mG (IQR = 5 mG) during acute concussion were found. All other timing intervals were not significantly altered.

Discussion

This is the first study to describe the cardiac cycle timing and contractility parameters in acute SRC. A significant decrease in systole was the only difference in timing intervals from baseline to the acute concussion phase (Days 1–5). However, the contractile parameter AS was significantly increased during the acute phase. These results suggest that systolic timing is altered and furthermore imply that there is excessive stress on the atrium during concussion. However, considering that IVRT was not significant, the elevation in AS is likely not due to restrictive filling, and the MVO to E-wave data suggests that it is not a result of elevated atrial pressure. Thus, we speculate here that during SRC, there may be an increased difficulty in the myocytes ability to maintain adequate pressure to overcome the mitral valve, thereby resulting in an increased AS, which consequently leads to an elevation in atrial stress and reduced systolic time.

Electrocardiography (ECG) has typically been used to evaluate cardiac function in athletic populations (Maron et al., 2007). However, with ECG only, there is a potential to miss any functional abnormalities. Furthermore, cardiac cycle timing data on concussed athletes are non-existent. In isolated mTBI, 19.6% of the patients showed impaired echocardiography data at 3-, 7-, and 14 days post-injury, including left ventricular ejection fraction or regional wall motion abnormality (Prathep et al., 2014). However, other research is also available to show that no systolic dysfunction was present 1 day following

Table 1. Data presented as median (interquartile range). IVRT = Isovolumic relaxation time; IVCT = Isovolumic contraction time; MVO to E-wave = Mitral valve open to E wave; REP = Rapid Ejection Period; AS to MVC = Atrial systole to mitral valve closure; DPI = Diastolic performance index (IVRT/systole); SPI = Systolic performance index (IVCT/systole); HPI = Heart performance index [(IVRT +IVCT)/systole]; TF = Twist force. *p<0.05.

Parameters	Baseline (N = 12)	Acute Injury (Days 1 to 5; N = 12)	Z-statistic	Significance
Age (years)		21 ± 2	–	–
Body Mass Index (kg/m^2)		26 ± 3	–	–
%Female		25	–	–
Heart Rate (bpm)	63 (11)	63 (9)	−1.17	0.239
IVRT (ms)	88 (10)	90 (17)	−0.71	0.48
IVCT (ms)	34 (7)	33 (6)	−0.16	0.875
Systole (ms)*	298 (42)	276 (32)	−2.12	0.034
Diastole (ms)	537 (202)	578 (186)	−0.16	0.875
MVO to E-wave (ms)	51 (9)	50 (15)	−0.63	0.53
Rapid Ejection Period (ms)	89 (28)	81 (18)	−0.94	0.347
AS to MVC (ms)	20 (8)	22 (8)	−0.86	0.388
DPI	0.32 (0.06)	0.33 (0.03)	−1.33	0.184
SPI	0.12 (0.05)	0.12 (0.02)	−0.63	0.528
HPI	0.44 (0.11)	0.45 (0.05)	−1.38	0.167
TF (mG)	18 (10)	23 (14)	−0.86	0.388
Atrial Systole (mG)*	10 (6)	13 (5)	−2.47	0.013

mTBI (Krishnamoorthy et al., 2017), although this was not due to SRC. Our results suggest that systolic time is decreased in the acute phase of SRC, which therefore does suggest systolic impairments and can imply subtle, subclinical systolic dysfunction. Indeed, maintenance of the myocyte LV pressure decreases, which results in a reduced systolic period (Biering-Sorensen et al., 2015). Therefore, we used a simple, non-invasive method to detect cardiac cycle timing intervals and contractile function. This method takes 1 minute to complete per participant and allows for observation of every heartbeat within the 1-minute period. Furthermore, considering the contrasting findings in the literature pertaining to SRC and HRV (Charron et al., 2021), a method to rapidly provide greater insight into the entire cardiac cycle can deliver a less ambiguous assessment.

The identification of atrial contractility dysfunction as demonstrated in our study provides unique insight into the mechanism associated with SRC. While reduced HRV and BRS parameters suggest that there is an elevation in sympathetic tone (S. A. Bishop et al., 2018), these parameters are associated with stimulation of the vagus nerve, thus influencing the sinoatrial (SA) and atrioventricular (AV) nodes, and can provide insights related to stress or inflammation (Kim et al., 2018; Williams et al., 2019). Previous research has demonstrated autonomic dysregulation during concussion (S. Bishop et al., 2017; Esterov & Greenwald, 2017), and thus we speculate based on our current findings that the nodal tissue within the right atrium is not able to function normally, or that the messages from the brain to the RA nodal tissue (SA or AV nodes) are disturbed.

When observing severe TBI patients within 24 hours, speckle tracking echocardiography has noted trends suggesting that systolic strain increases in the TBI state (Cuisinier et al., 2016). Elevated strain parameters imply greater myocardial deformation and represent elevated contractility. Our results suggested that there is an elevated atrial contraction contribution to ventricular filling, and while diastolic contribution is not significantly altered, some of the changes in ventricular filling may be due to the shortened systolic time. Changes in vascular responses have been noted in concussion, such as reduced cerebral blood flow (Giza & Hovda, 2014; Len & Neary, 2011). In more severe vascular brain injury, ECG left atrial abnormalities such as P-wave terminal force are associated with vascular brain injury (Kamel et al., 2015); however, there is a distinct lack of information associating changes in atrial function to those with concussion. Therefore, our results are the first to suggest that concussion can alter atrial contractile function within 5 days post-injury (Figure 1).

Research in acute concussion has emphasized changes in cerebrovascular function, with altered cerebral blood flow (CBF) and cerebral autoregulation known to be common impairments (Giza & Hovda, 2014; Len & Neary, 2011; Wright et al., 2018). CBF decreases during the acute stage, suggesting altered ionic and metabolic activity (Champagne et al., 2021; Giza & Hovda, 2014). This can be reflective of the vasoreactivity which can occur due to acute concussed (Aaron et al., 2021), which along with the CBF decrease can lead to a mismatch between metabolic requirement in comparison to the perfusion during the CBF perturbations (Giza & Hovda, 2014). The perfusion impairments may be due to a reduction in systole in the presence of sustained HR, thereby limiting systemic circulation and altering cerebral haemodynamics during the acute concussion period. The reduction in systolic time can also explain why some patients with isolated TBI exhibit a reduced left ventricular ejection fraction, as there may be limitations in ensuring

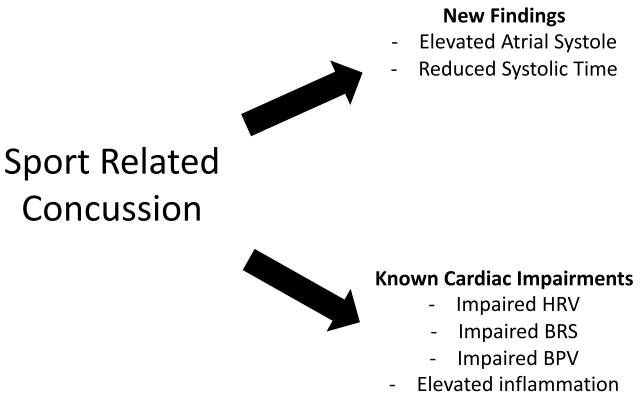

Figure 1. Cardiac dysfunction following sport-related concussion include impairments in heart rate variability (HRV), baroreflex sensitivity (BRS), and blood pressure variability (BPV). New group findings include depressed systolic time and elevated atrial stress.

adequate stroke volume under resting conditions. This reduction in stroke volume (likely due to reduced systolic time) can further be suggestive of a potential cardiovascular dysfunction sequelae which occurs following concussion.

Limitations include the generalizability of our study results being applied to a younger, athletic population. The athletes in this study varied in their strength (American football), and endurance (ice hockey) training which can impact cardiac function. However, we did use a within-subject analyses to reduce the influence of sport and the nature of training. Finally, the concussion data is limited by a small sample size with both female (n = 3) and male (n = 9) varsity-level athletes.

Conclusion

This study supports existing research that cardiac dysfunction occurs during acute concussion but provides novel insights into the uniqueness of cardiac cycle timing and contractility dysfunction resulting from concussion, which can vary from athlete-to-athlete. These results support the need for baseline testing for all varsity level or competitive athletes and provide further evidence that a personalized approach for cardiac recovery in athletes with SRC can be beneficial. We report novel changes in AS and systolic time implying that cardiac alterations must be assessed when treating a concussion. Finally, this study provided new evidence that the atria or issues related to atrial contraction can be impaired following a sport-related concussive injury. More research is warranted in this area.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

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